

unmanufacturability” published on 21st April 2011 in IOP Publishing’s journal Nanotechnology argues that you cannot mass produce structures with a dia of three nanometres or less using a top-down approach.

The overall goal when entering nanotechnologies into the market is low-cost, high-volume manufacturability, but at the same time, the materials’ properties must be highly reproducible within a pre-specified limit, which Kelly states cannot happen below the 3nm limit when trying to make arrays. According to Kelly, it doesn’t matter if even the most modern forms of deposition (including epitaxy), e-beam or ultra-deep UV lithography and precision etching, the mainstay of microelectronics and optoelectronics fabrication, is being used; there are strict limits described below for which one-off fabrication is possible, but manufacture is not.

The top-down approach to manufacturing, which Kelly states is limited, uses external tools to cut and shape large materials to contain many smaller features. Its alternative, the bottom-up approach, involves piecing together small units, usually molecules, to construct whole materials – much like a jigsaw puzzle – however this process is too unpredictable for defect – free mass production of arrays.

Kelly used statistical evaluation of vertical nanopillars - that have been suggested for uses in sensors and displays - as an example to demonstrate his theory. He states that the proof comes in two stages. The first is due to the fact that when materials are mass produced on such a small scale there will be a lot of variation in the size of different components. As a result of this variation, the properties of the material will vary to an extent where the material cannot function

He refers to the paper “Nanowires: fabrication of Sub-10 nm metallic lines of low line-width roughness by hydrogen reduction of patterned metal-organic materials” which deals with quantum wires at 7 nm design rules showing that the conditions on uniformity within structures can be achieved that just reaches the levels required for silicon integrated circuit wiring. He concludes that the interface between manufacturability and unmanufacturability must be somewhere in between 3 and 7 nm. For silicon CMOS technology, the limit is higher, since reliable capacitors cannot be built smaller than the 10-15nm range.

However this paper doesn’t address any physical limitations related to “bottom-up” manufacturing approaches, such as chemical or biological based molecular synthesis, which is another area of research for mass producing nanotechnology.

Professor Kelly says, “If I am wrong, and a counterexample to my theorem is provided, many scientists would be more secure in their continued working, and that is good for science.”